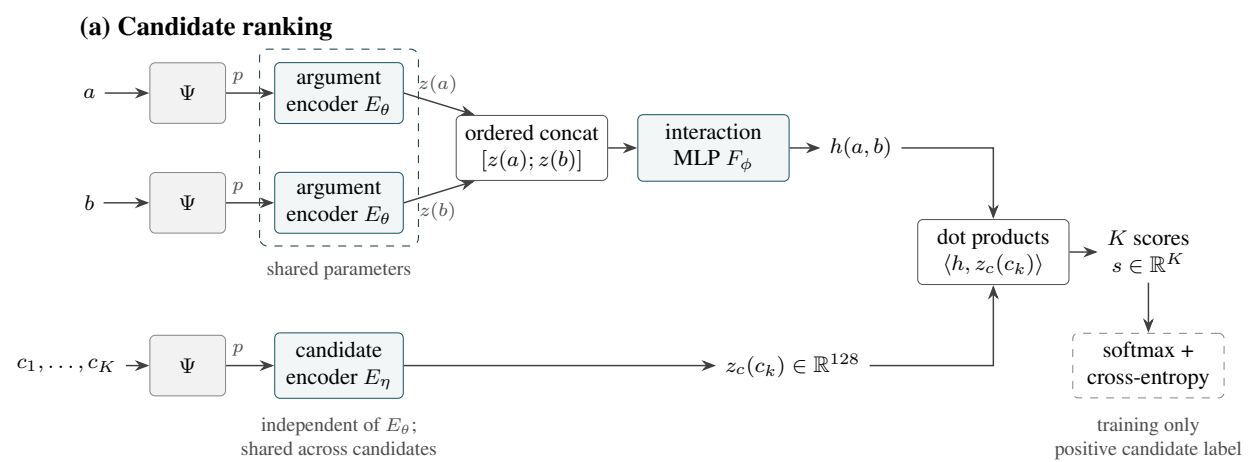

(a) Candidate ranking
a
Ψ
p
argument encoder E_θ
$z(a)$
b
Ψ
p
argument encoder E_θ
$z(b)$
shared parameters
ordered concat $[z(a); z(b)]$
interaction MLP F_ϕ
$h(a, b)$
dot products $\langle h, z_c(c_k) \rangle$
K scores $s \in \mathbb{R}^K$
softmax + cross-entropy
training only positive candidate label
$c_1, \ldots, c_K$
Ψ
p
candidate encoder E_η
$z_c(c_k) \in \mathbb{R}^{128}$
independent of E_θ; shared across candidates
(b) Layers inside each learned module
Argument encoder E_θ
affine: $p \to 256$
GELU
affine: $256 \to 128$
Interaction MLP F_ϕ
affine: $256 \to 512$
GELU
affine: $512 \to 128$
Candidate encoder E_η
affine: $p \to 256$
GELU
affine: $256 \to 128$